

Operations Initial Funding Request Form

Approved by EPAC Chairmen external to meeting on 05/09/2023

[Link to 05/11/2023 EPAC Meeting Minutes](#)

Project Title: Hyde Park BESS – Battery Energy Storage System Solution, Rev 1	Project Number: 21334
Date Prepared: 06/10/2021, 03/21/2023 Rev 1	Company: Eversource EMA
Organization: System Planning	Class(es) of Plant: Other
Project Initiator: Ling Yang	Project Category: Stations - Batteries
Managing Organization: Engineering	Schedule ID#: TBD
Project Manager: Rosenberg, Michael P	Project Type: Specific
Project Sponsor: Freeman, Lavelle A	Capital Investment part of original Op Plan: Yes
Estimated in service date: Dec 2025	Emergency Related Request? No
Eng./Constr. Resources Budgeted? Yes	Facility Type (check all that apply): ☐ PTF ☒ non-PTF ☒ Distribution ☐ N/A (General Plant)
Total Capital Request: \$300K	Tx Development Costs: No

[1] PROJECT NEED STATEMENT

REV 1 – This is a revised initial funding request of \$300K to fund initial engineering studies for installation of a 10 MW/20 MWh utility-scale battery storage system (BESS) along Cummings Highway near Hyde Park Substation #496 in Hyde Park, MA. See attached Figure 1 for the geographic layout of the Hyde Park area and the proposed BESS location and Figure 2 for details.

The main proposed project is to help relieve the heavy loading condition of Hyde Park Substation #496. In the future, the battery system could be used to test a new concept of installing a BESS connected to multiple voltage levels, moving energy between Hyde Park #496 and Baker Street #110 (located in West Roxbury, MA) substations with different voltage levels and test the charge and discharge of the battery system at the same time.

Hyde Park substation #496 is heavily loaded above 95% of the substation load carrying capability (LCC). A near-term load transfer and a mid-term solution to bring additional capacity to the area is needed before a long-term solution (21192 - Hyde Park-Dorchester Area Supply Long-Term Plan, IFR was approved in May 2021) will be in service. This proposed BESS is the proposed mid-term solution.

The proposed BESS installation will not only serve as the mid-term solution to bring the Hyde Park substation loading below 95% of the substation firm LCC but can potentially test new BESS applications connected at different voltage level.

Note: The original IFR was approved in PowerPlan for \$200K on June 16, 2021, sponsored under Grid Mod program. Since the Hyde Park station #496 was overloaded in 2021, leadership decided to move this project out of Grid Mod and proceed under the capital project track. As of March 27, 2023, \$169K has been spent on engineering evaluation and project management.

The revised amount of \$300K will support the project through presentation at the Solution Design Committee (SDC).

[2] PROJECT OBJECTIVES

The primarily objective of the project is to increase the Hyde Park area capacity and relieve the heavy loading condition of the station. The BESS would serve as a mid-term solution for Hyde Park before the area long-term solution can be in place.

The project potentially can be used in the future to test other new applications/usages of the battery storage system. The project could demonstrate new BESS design/application ideas by connecting BESS system to various system voltage levels, (14kV and 25kV distribution systems); moving energy from one (1) substation to the other when two (2) systems cannot be directly connected; and testing partial discharging of the BESS while the remaining portion is charging with grid.

See attached Figure 3.1 for a simplified one-line diagram and Figure 3.2 for the one-line diagram with potential future connection to Baker Street substation. See attached Figure 4 for the area feeder diagram and potential BESS site.

The benefits of the proposed design are as follows (Rev. 1):

- Bring additional capacity to Hyde Park #496
- Reduce the need to perform break before make emergency load switching
- Serve as the mid-term solution to prevent substation from overload
- Keep transformer cooler for pre-contingency event

In the future, the project potentially can be connected to 25kV side, so it could be:

- Pilot project to test connecting BESS to multi-level voltage systems
- Test moving energy through BESS between substations from one (1) to the other when two (2) systems cannot be directly connected

- Test partial BESS charging with grid while the remaining portion is discharging, which could potentially reduce the MWh needed compared to a system that cannot charge/discharge simultaneously.
- Avoid load profile being flat at a heavily loaded substation
- Provide additional capacity to Baker Street #110 during equipment maintenance in the near-term and load relief in the long-term

Alternatives:

- One alternative to be evaluated as part of this project is to construct new distribution lines from Needham Station#148, located in Needham, MA to bring an additional capacity to Hyde Park Station #496.
- Stage a mobile transformer in the area to increase area capacity.

[3] FUNDING REQUEST EXPLANATION (total request, amount per task, deliverables)

A detailed project scope including quantities, materials, manufacturer selection, has not been prepared yet, and detailed feedback from all affected disciplines (Station Engineering, Control and Protection, Communication Engineering, Environmental, etc.) has not yet been attained as part of this project. This \$300K initial funding request will be used to develop preliminary engineering, including the following:

Station Engineering - \$65K

Provide station design scope and cost estimates for potential solutions

System Planning – \$50K

Perform system analysis and study to evaluate the system impacts

Distribution Engineering--\$65K

Evaluate street work feasibility, scopes and provide cost estimates for solutions.

Evaluate street work feasibility, scopes, and cost estimates to bring an additional 10MVA capacity to Hyde Park area by constructing new distribution lines from Needham Station and determine when distribution work becomes cost prohibited.

Environmental and Siting - \$25K

Identify environmental and siting requirements for potential solutions

Project Management - \$40K

Collaborate engineering team works

Control and Protection - \$30K

Identify control and protection upgrade requirements for potential designs

Communication Engineering – \$25K

Identify communication scope/cost for potential designs

It is anticipated that the project will be presented to Solution Design Committee (SDC) in September 2023 and to EPAC for partial funding in March 2024.

[4] INITIAL FUNDING SCHEDULE

Milestone	Scheduled Completion Date
SDC Approval (for applicable projects)	September 2023
Conversion of Tx development costs, 183 to 107 (applicable to projects that require SSF approval and/or ISO preferred alternative selection)	N/A
EPAC Partial Funding Approval (for applicable projects)	March 2024
EPAC Full Funding Approval	September 2024
Start of Construction	October 2024
Project In-Service Date (ISD)	December 2025
Project Need Date (for applicable projects)	Present

[5] REFERENCES

References
1. 21192 -Hyde Park -Dorchester Area Plan IFR
2.
3.

[6] ATTACHMENTS (One-Line Diagrams, Images, etc.)

Attachment
Figure 1 - Geographic Layout of the Hyde Park Area
Figure 2 - Geographic Layout of the Hyde Park Area (Detail)
Figure 3.1 - Simplified One-Line Diagram
Figure 3.2 - Simplified One-Line Diagram with potential future connection to Baker Street side
Figure 4 - Area Feeder Diagram and potential BESS site

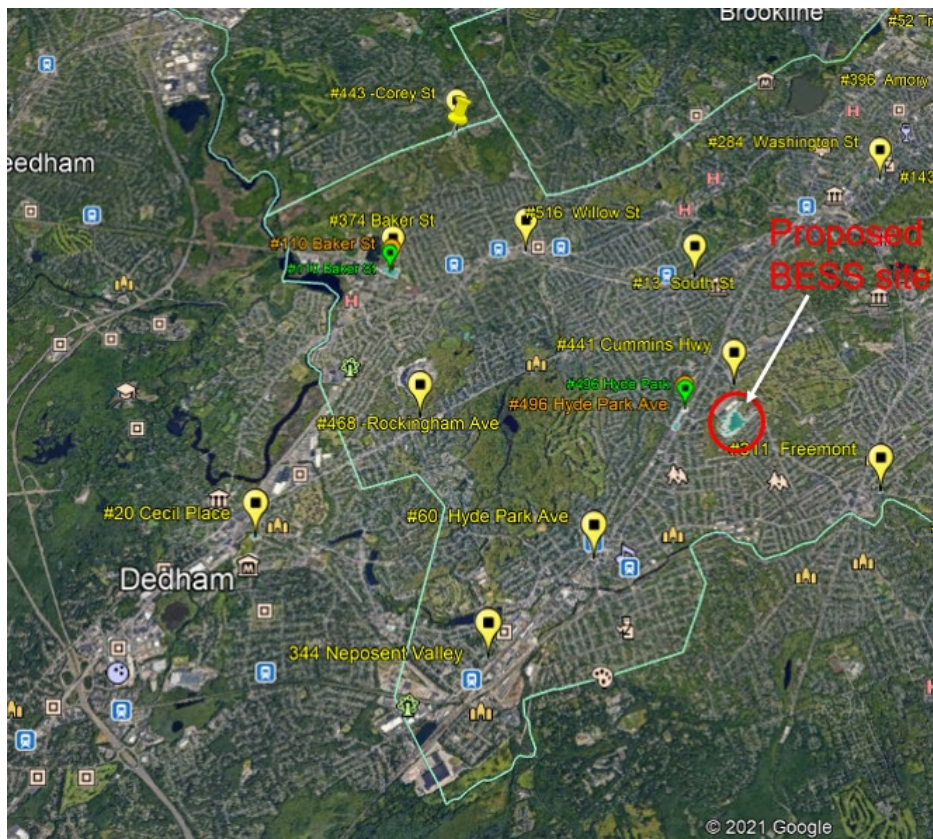


Figure 1 - Geographic layout in the vicinity of Hyde Park and Proposed BESS site



Figure 2 - Geographic layout in the vicinity of Hyde Park and Proposed BESS site (zoom in)

Hyde Park Proposed BESS One-Line Diagram

Substation, Feeder, and Equipment Data

EVERSOURCE
ENERGY

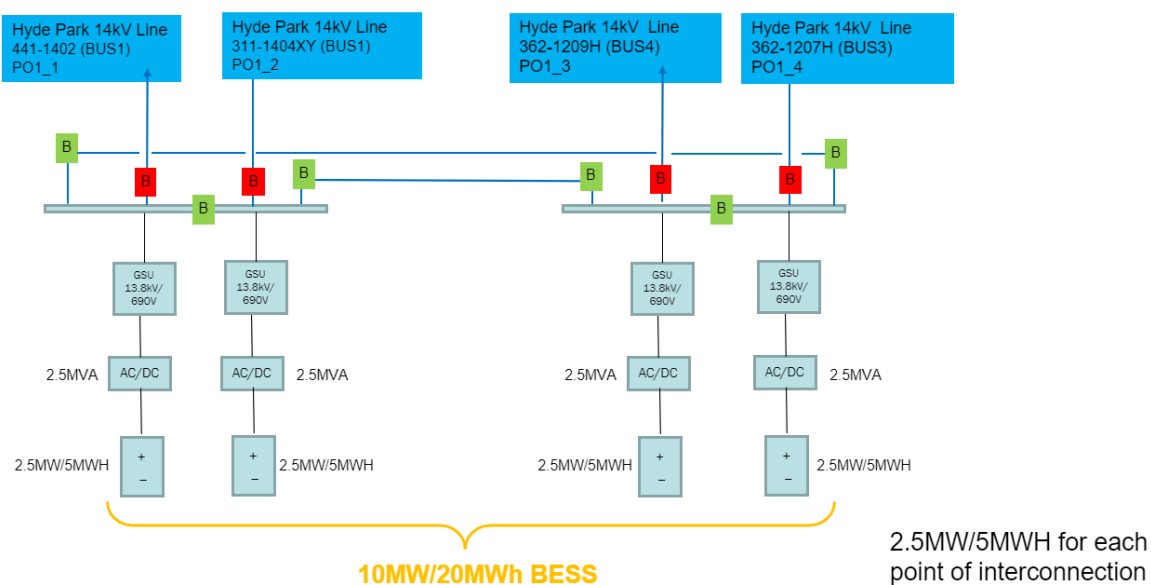


Figure 3.1 - Simplified One-Line Diagram

Potential future connection to Baker St One-Line Diagram

EVERSOURCE
ENERGY

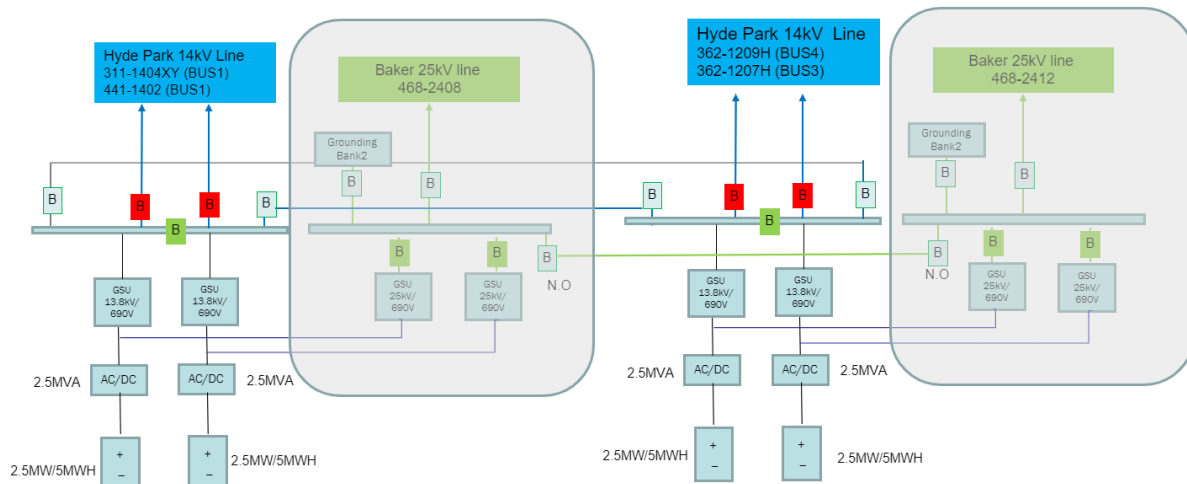


Figure 3.2 - Simplified One-Line Diagram with potential future connection to Baker Street side (shaded area)

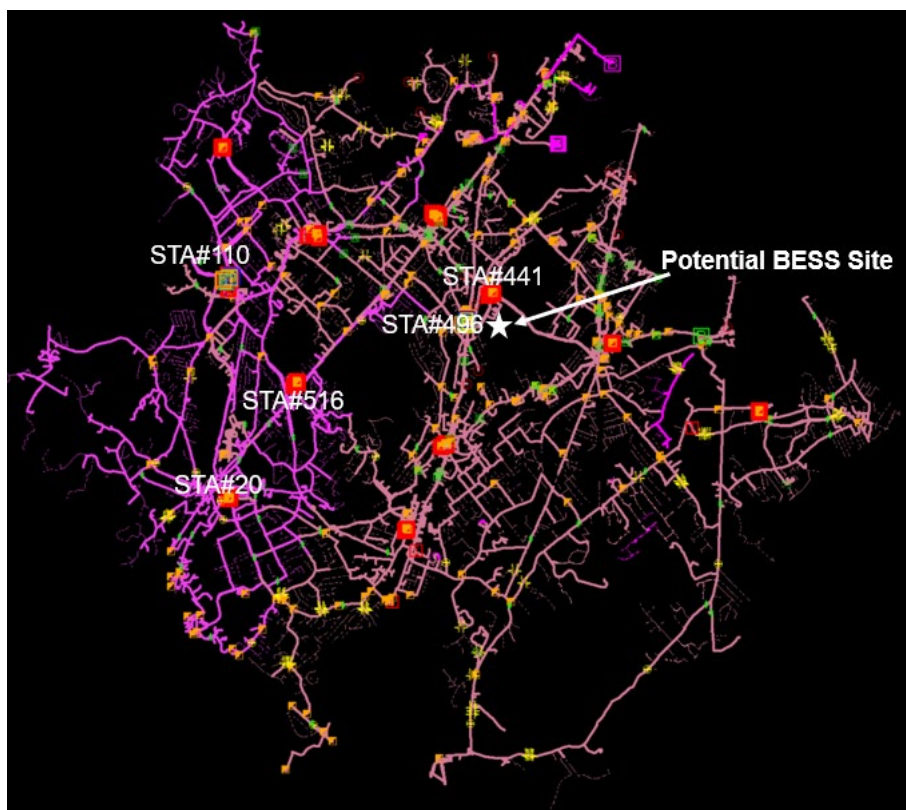


Figure 4 - Area Feeder Diagram and potential BESS site